This is a README file that explains the protocol followed to generate the dataset using two IMU sensors mounted on the feet.

1 Collected data

The collected data include raw IMU data, such as acceleration, quaternions, etc. In addition, seven generated features from the Kalman filter are recorded: the horizontal and vertical positions of the swing foot, the inclination angles of both feet, and the angle formed between the horizontal and vertical positions in the sagittal plane. Gait subphases and phases of both feet are also recorded. The following table summarizes all the 28 data considered during the collection process:

No	Variable name	Data	Unit
1	index	Sample counter (100Hz)	N/A
2	x_acc_left	Acceleration x-axis on left foot IMU	m/s^2
3	y_acc_left	Acceleration y-axis on left foot IMU	m/s^2
4	z_acc_left	Acceleration z-axis on left foot IMU	m/s^2
5	x_acc_right	Acceleration x-axis on right foot IMU	m/s^2
6	y_acc_right	Acceleration y-axis on right foot IMU	m/s^2
7	z_acc_right	Acceleration z-axis on right foot IMU	m/s^2
8	quat_1_left	Orientation quaternion(1) on left foot IMU	Unit quaternion
9	quat_2_left	Orientation quaternion(2) on left foot IMU	Unit quaternion
10	quat_3_left	Orientation quaternion(3) on left foot IMU	Unit quaternion
11	quat_4_left	Orientation quaternion(4) on left foot IMU	Unit quaternion
12	quat_1_right	Orientation quaternion(1) on right foot IMU	Unit quaternion
13	quat_2_right	Orientation quaternion(2) on right foot IMU	Unit quaternion
14	quat_3_right	Orientation quaternion(3) on right foot IMU	Unit quaternion
15	quat_4_right	Orientation quaternion(4) on right foot IMU	Unit quaternion
16	x_gyro_left	Angular rate x-axis on left foot IMU	rad/s
17	y_gyro_left	Angular rate y-axis on left foot IMU	rad/s
18	z_gyro_left	Angular rate z-axis on left foot IMU	rad/s
19	x_gyro_right	Angular rate x-axis on right foot IMU	rad/s
20	y_gyro_right	Angular rate y-axis on right foot IMU	rad/s
21	z_gyro_right	Angular rate z-axis on right foot IMU	rad/s
22	Feature1	Absolute horizontal position feature	m
23	Feature2	Vertical position feature	m
24	Feature3	Absolute horizontal velocity feature	m
25	Feature4	Vertical velocity feature	m
26	Feature5	Direction angle of velocity vector	Deg.
27	Feature6	Angle of non-reference (swing) foot	Deg.
28	Feature7	Angle of reference (stance) foot	Deg.
29	label	Actual gait mode label	N/A
		(1: level walking, 2: ramp up, 3: ramp down, 4: stair up, and 5: stair down)	

Table 1: Nature of the collected data

2 Experimental Procedure

The protocol involves collecting kinematic walking data from several subjects as they walk in various environments. Our main objective is to capture the transitions between these environments. In total, we have eight daily living transitions: level walking to ramp (ascent/descent), level walking (LW) to stairs (ascent/descent—SA/SD), ramp (ascent/descent—RA/RD) to level walking, and stairs (ascent/descent) to level walking. These transitions are represented as follows: LW $\leftrightarrow$ SA, LW $\leftrightarrow$ SD, LW $\leftrightarrow$ RA, LW $\leftrightarrow$ RD.

1. **Sensor Setup:** The subject is asked to wear two IMU sensors on his/her feet. A calibration process is conducted to set the angles to their rest position (0 degrees) and to initialize the Kalman filter to prevent drift during walking.

2. **Walking Procedure:** For each subject, a total of seven steps are recorded. The subject is instructed to begin with their right foot to ensure coherence and consistency in the data. The first and seventh steps are considered half-steps, as the walk begins and ends with these. These two half-steps are excluded during data processing.

3. **Transition Steps:** The transition occurs at the fourth and fifth steps, with the transition spanning two steps. The preceding steps are classified as the "late gait mode," while the steps after the transition are classified as the "new gait mode."

4. **Repetitions:** Each transition is performed a minimum of eight times for each subject.

2.1 Number of participants:

A total of five subjects participated in the data collection. They were briefed on the experimental procedure and provided written consent before participating. The following table summarizes their demographic information:

Participants	Age	Weight (KG)	Height (m)
1 (M)	26	73	1.85
2 (M)	26	110	1.92
3 (F)	23	63	1.68
4 (M)	24	87	1.78
5 (F)	35	68	1.74

M and F stand for male and female, respectively.

3 Repetition and averaging of the data

Once the experiment for each transition is completed, the collected data are averaged to assess the consistency of the repetitions. For example, the following figure shows the averaged five steps of the transition from level walking (LW) to stair ascent (SA):

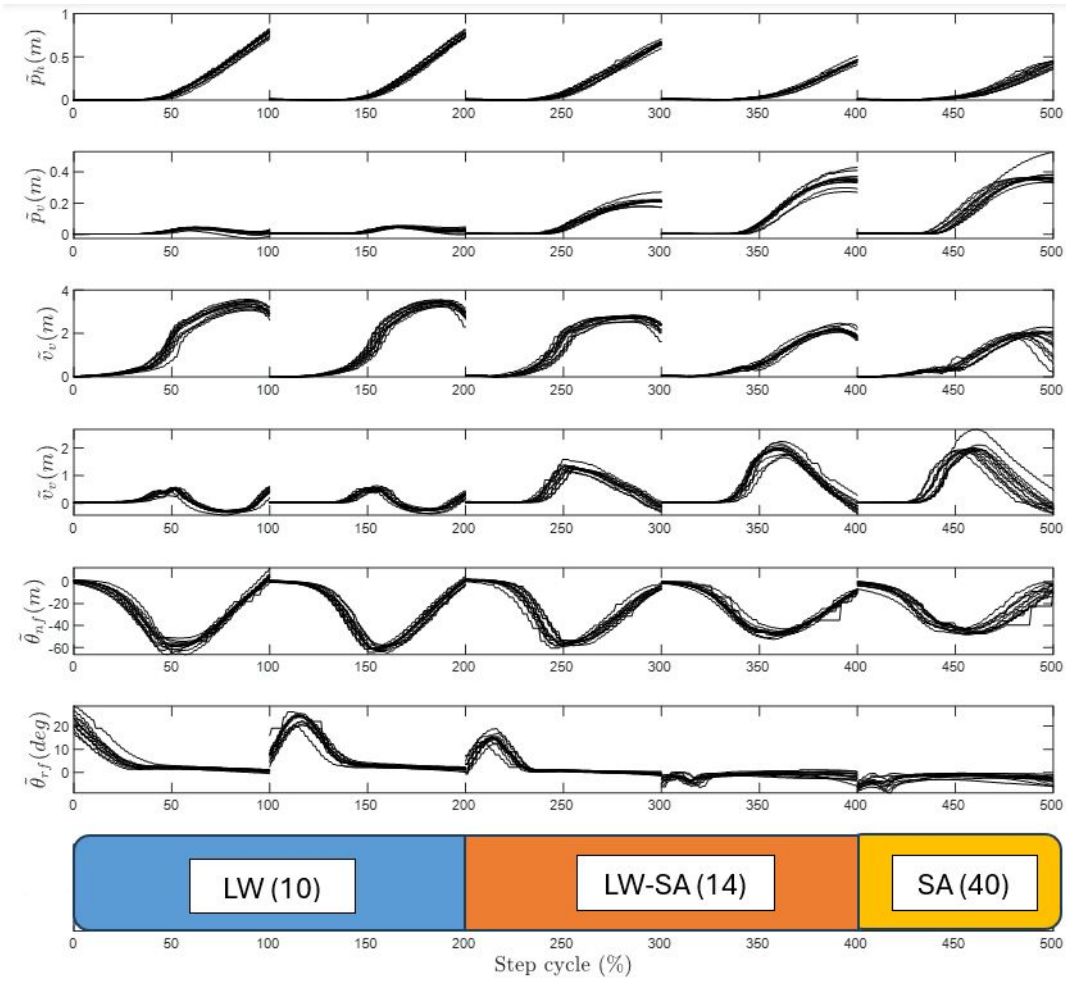


Figure 1: Evolution of the kinematic feautres over five steps

One can clearly observe that all the signals are well synchronized at each step and no overlap is happening which shows a consistency of the data

Data Labeling

- The labeling of the gait modes is structured such that each gait mode is represented by two digits:
  - LW = 10
  - RA = 20
  - RD = 30
  - SA = 40
  - SD = 50

The "0" after the first digit indicates that the walking remains continuous within the current environment (i.e., no transition has occurred).

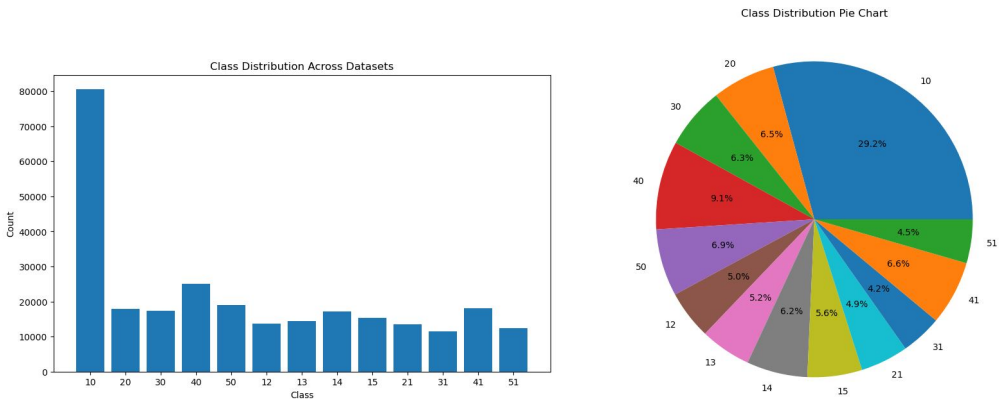
- Transitions are labeled using the following notation:
  - LW → RA: 12
  - LW → RD: 13
  - LW → SA: 14

- LW → SD: 15
- RA → LW: 21
- RD → LW: 31
- SA → LW: 41
- SD → LW: 51

For transition labeling, the first digit represents the previous gait mode, and the second digit represents the next gait mode.

4 Distribution of the data samples

Overall, we have 8 transitions plus 5 gait modes, which results in a total of 13 classes. Here follows some statistical plotting of the data per each class as well as the number of samples. The following figure shows the number of samples per each class:



(a) (b)

Figure 2: Statistical analysis of the data as a function of the number of samples per subject/class